41_Lumina

Domain : ‘Education and Academic Administration’

Problem Statement - ‘Course Enrollment And Academic Planning System’

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Preface

Course Enrollment and Academic Planning System provides a clear and shared understanding of how academic enrollment and planning activities are expected to function within an institution. The document is intended for students, faculty and academic deans, and the technical team involved in designing and implementing the system.

The system replaces fragmented and manual enrollment practices with a centralized, rule-based platform. Manual methods such as spreadsheets or online forms often lead to errors, delays, and inconsistent academic data. The system automates enrollment and academic planning according to predefined institutional rules.

Key academic concepts—such as **Course**, **Course Credit**, **Enrollment**, **Credit Requirements**, **Core Course**, **Degree Requirements**, **Tentative Academic Roadmap**, and **Degree Audit**—are formalized to ensure consistent interpretation and application of academic regulations. It performs real-time validation of prerequisites, credit limits, seat availability, and schedule constraints.

The system supports five main actors—**Student**, **Faculty**, **Assistant Dean** 1 ,**Assistant Dean** 2 and **Dean** —with defined responsibilities. Students plan and enroll in courses, faculty manages academic content and student evaluation, Assistant Dean 1 allocates course slots, manages the course catalog, Assistant Dean 2 controls enrollment windows and grade sheets of students and Dean oversees and manages all actions of Assistant Deans.

This document serves as a reference for developers, reviewers, and academic stakeholders, providing a foundation for system design, implementation, and future improvements.

Introduction

The Course Enrollment and Academic Planning System is a centralized academic platform that supports structured course registration and long-term academic planning. It enables a **student** to enroll in courses based on predefined rules, track earned **Course Credits**, and plan future semesters through an **Tentative Academic Roadmap** aligned with institutional **Degree Requirements**. Every enrollment action is validated in real time to ensure compliance with prerequisites, credit limits, and curriculum rules.

The system also supports academic administration by providing role-specific capabilities to **faculty** and **Deans**. Faculty members manage course details, instructional schedules, and academic rosters, while Assistant Dean 1 maintains the course catalog, and ensures conflict-free scheduling Assistant Dean 2 configures enrollment policies and submits gradesheets. Through these roles, the system enforces **Credit Requirements**, preserves the sequencing of **Core Courses**, and maintains consistency across departments.

By integrating enrollment, planning, and academic progress verification into a single platform, the system acts as a reliable reference for monitoring progress through automated **Degree Audit** mechanisms. This structured approach reduces administrative effort, eliminates data inconsistency, and ensures that students progress toward graduation in a transparent and regulation-compliant manner.

Supporting workflows include enrollment window configuration, academic roster management, degree audits, and exception handling through override requests.The system enforces institutional academic rules automatically, minimizing manual intervention and error.

System Requirements

Functional Requirements :

User Authentication and Authorization :

- The system shall authenticate users based on their institutional credentials.

Enrollment Constraints And Configuration Window Management :

- The system shall allow the Assistant deans to define, update, and manage academic and enrollment policies, including the course catalog (courses, credits, prerequisites, core classifications), enrollment constraints, and enrollment windows.
- The system shall enforce enrollment constraints automatically, by blocking enrollments outside the active enrollment window, preventing scheduling conflicts such as overlapping course allocations, and supporting system-wide validation of eligibility.
- The system shall support administrative oversight and exception handling, by allowing deans to intervene in and resolve exceptional enrollment or scheduling conflicts and by providing a system-wide indicator showing whether the enrollment window is active or inactive.

Course Catalog Management :

- The system shall allow Assistant Dean 1 to create, update, and manage course details including course credits, prerequisites, and slot assignments.

- The system shall maintain a centralized course catalog accessible to students and faculty.

- The system shall ensure that prerequisites are consistently enforced across all related courses.

Course Enrollment Management:

- The system shall allow students to enroll in courses only during an active enrollment window.

- The system shall verify all Enrollment Constraints before confirming enrollment.

- The system shall automatically update the academic roster upon successful enrollment.

Prerequisite and Credit Validation :

- The system shall validate that all mandatory prerequisites are completed before allowing course enrollment.

- The system shall ensure that the student's total enrolled credits do not exceed institutional credit limits.

- The system shall block enrollment attempts that violate prerequisite or credit requirements and display appropriate error messages.

Schedule Conflict Prevention :

- The system shall prevent students from having overlapping course slots.

- The system shall notify the Assistant Dean 1 when a scheduling conflict occurs.

Academic Roadmap Generation :

- The system shall allow students to generate a personalized academic roadmap spanning multiple semesters.

- The system shall perform a degree audit to identify completed, pending, and remaining degree requirements.

- The system shall validate prerequisite dependencies while creating or modifying the roadmap.

- The system shall allow students to save and update their academic roadmap for future reference.

Override Request Handling :

- The system shall allow students to submit override requests when enrollment constraints cannot be met.

- The system shall allow the Assistant Dean 2 to approve or reject override requests.

- The system shall update enrollment status automatically upon approval of an override request.

Academic Roster Management :

- The system shall generate and maintain an official academic roster for each course.
- The system shall provide faculty members access to view enrolled student lists.
- The system shall ensure that rosters reflect only verified and validated enrollments.

Non Functional Requirements

Performance Requirements :

- The system shall process enrollment requests in real time with minimal latency.
- The system shall support concurrent enrollment attempts during peak enrollment periods.

Reliability and Availability :

- The system shall ensure accurate enforcement of **Credit Requirements** and **Degree Requirements** at all times.
- The system shall ensure transactional consistency during high-load enrollment scenarios.

Security Requirements :

- The system shall protect sensitive academic and personal data through secure authentication mechanisms.
- The system shall enforce role based access control for students, faculty and deans.
- The system shall maintain audit logs for critical operations such as enrollment approvals and overrides , to ensure accountability, and resolve disputes in the academic system.

Usability Requirements :

- The system shall provide a clear and intuitive user interface for students, faculty, and deans.

- The system shall display meaningful error messages when enrollment constraints are violated.

Scalability :

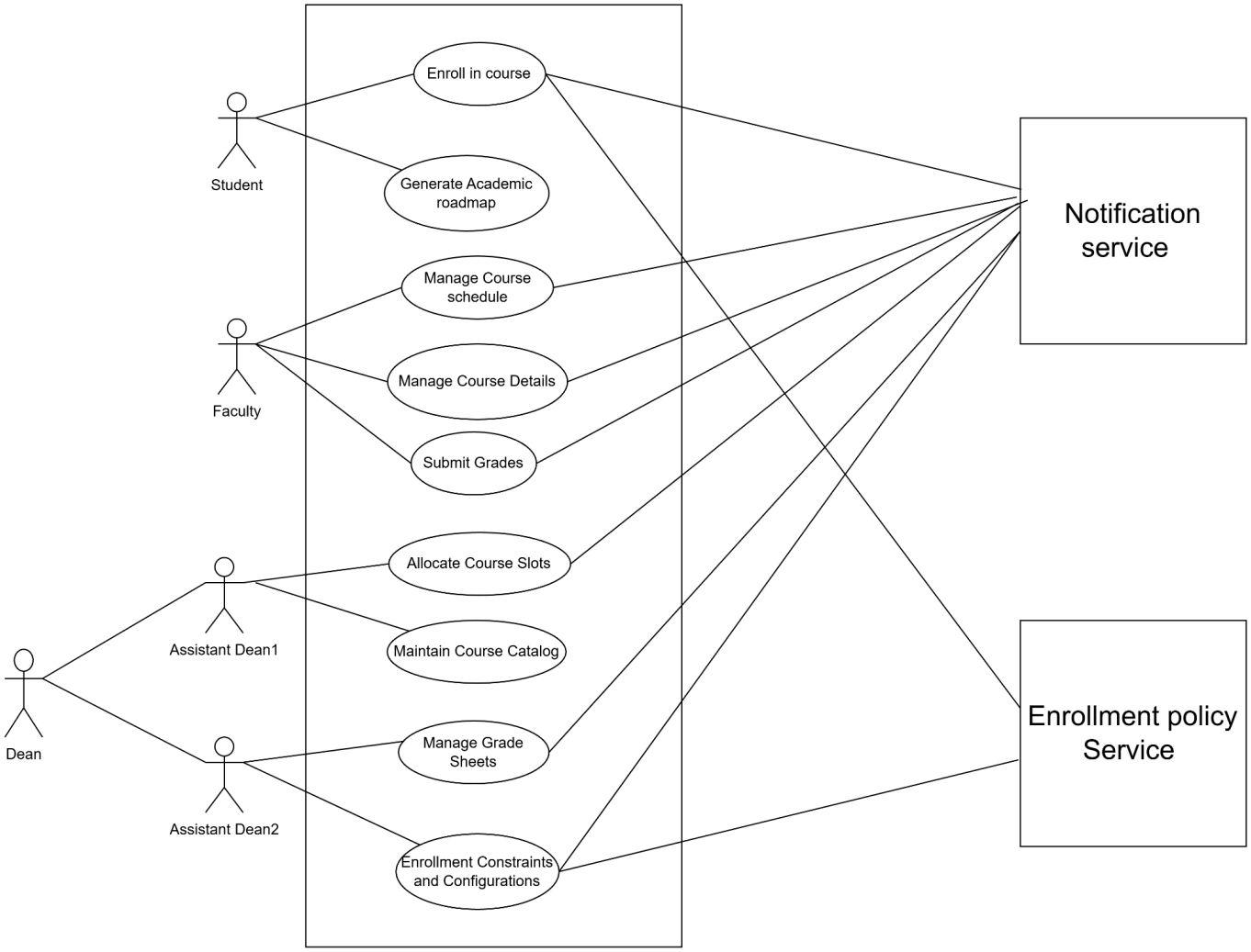
- The system shall scale to support increasing numbers of users and courses without performance degradation. The system shall handle peak enrollment loads during registration windows.

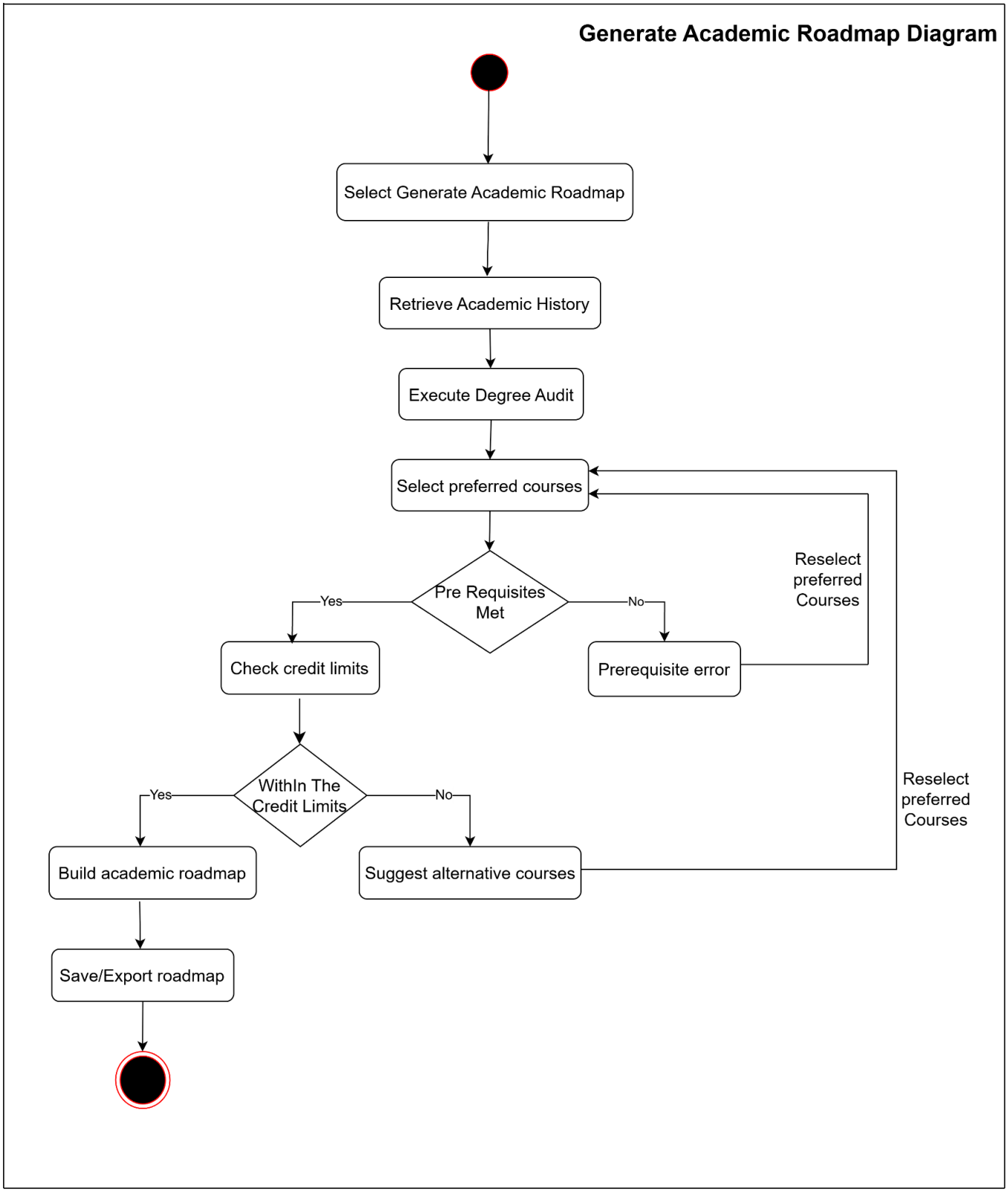
- The system shall enforce all institutional academic policies without manual intervention.

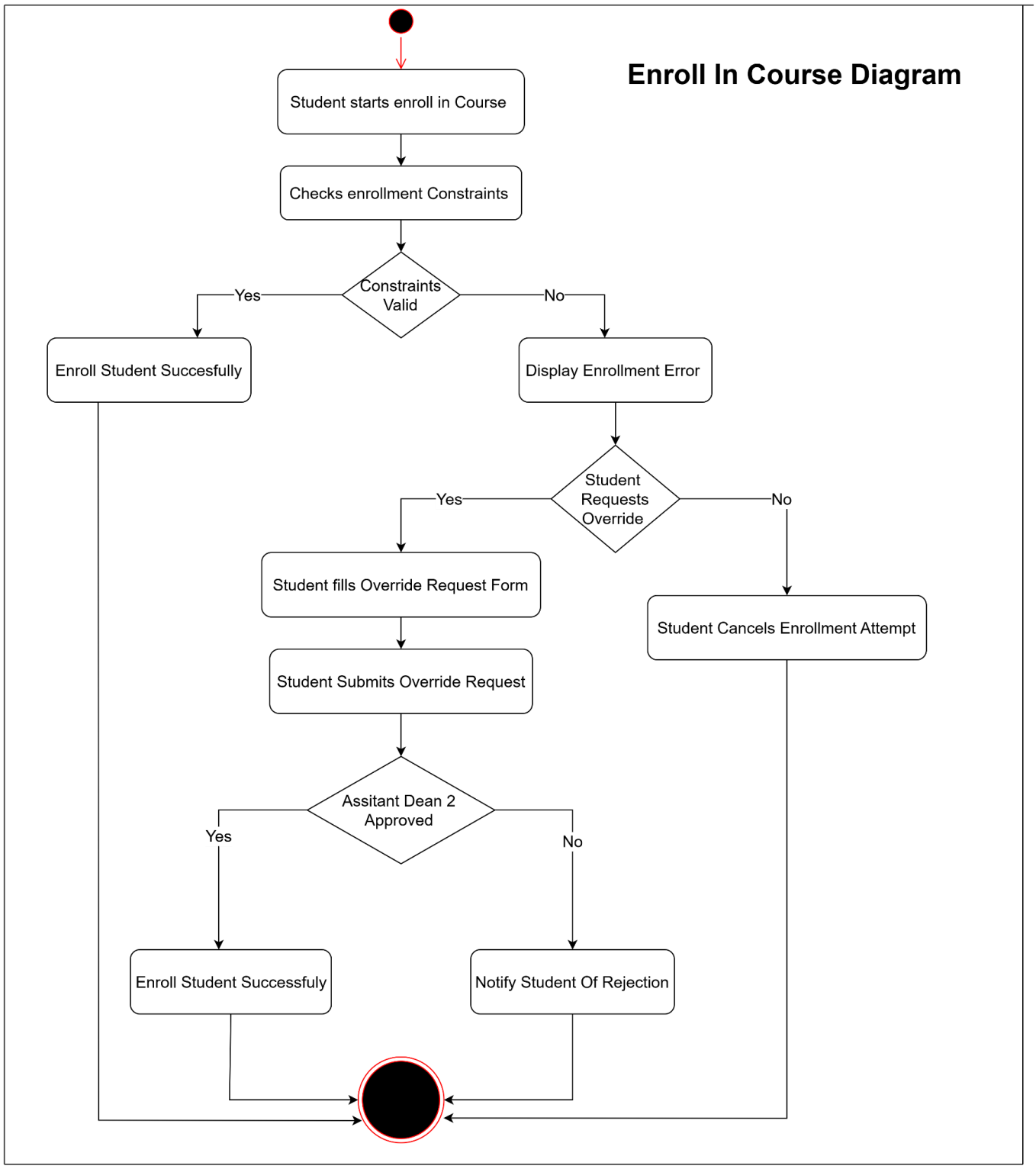
- The system shall allow enrollment only within authorized enrollment windows.

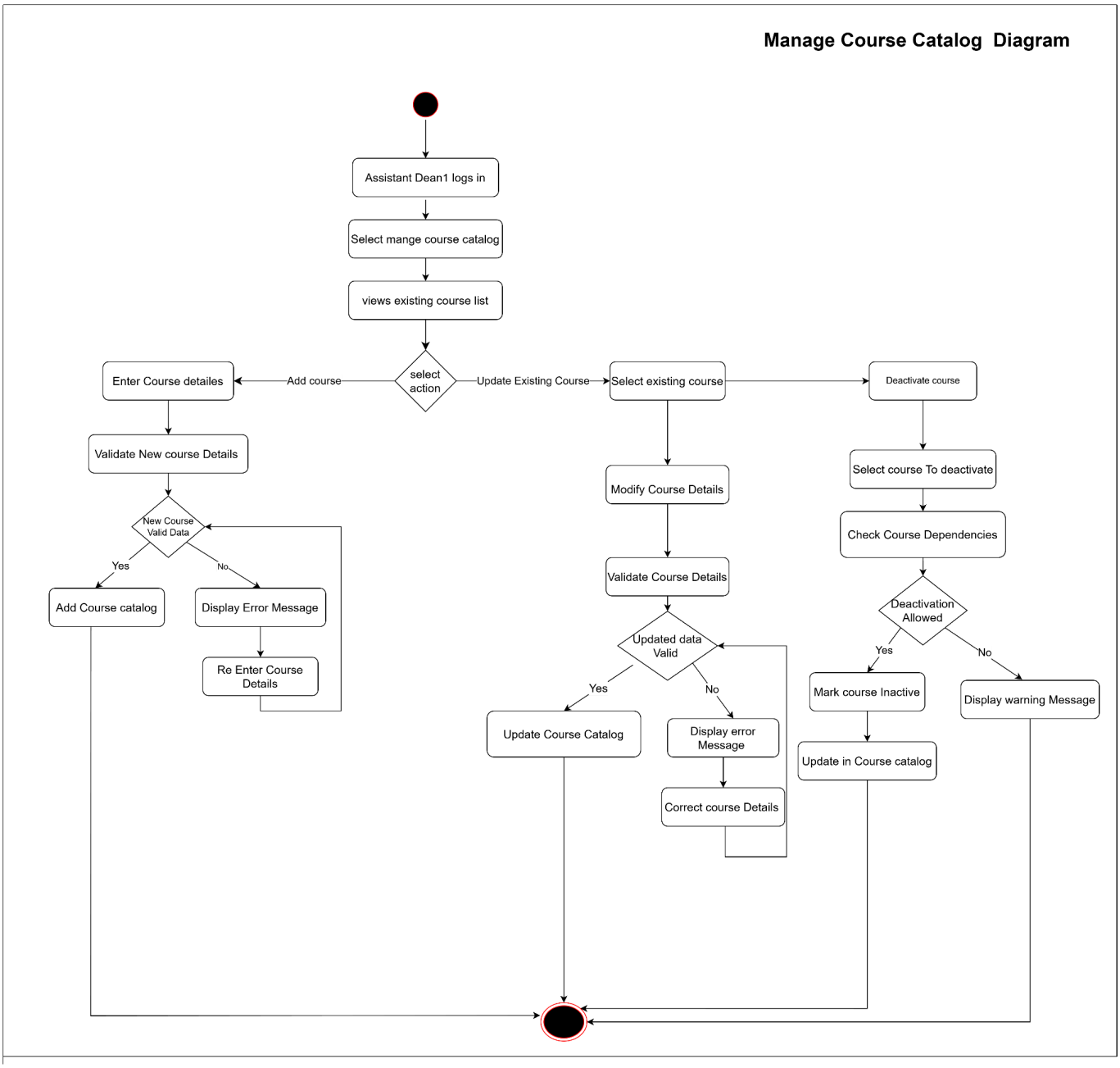
- The system shall strictly adhere to defined prerequisite and credit requirement rules.

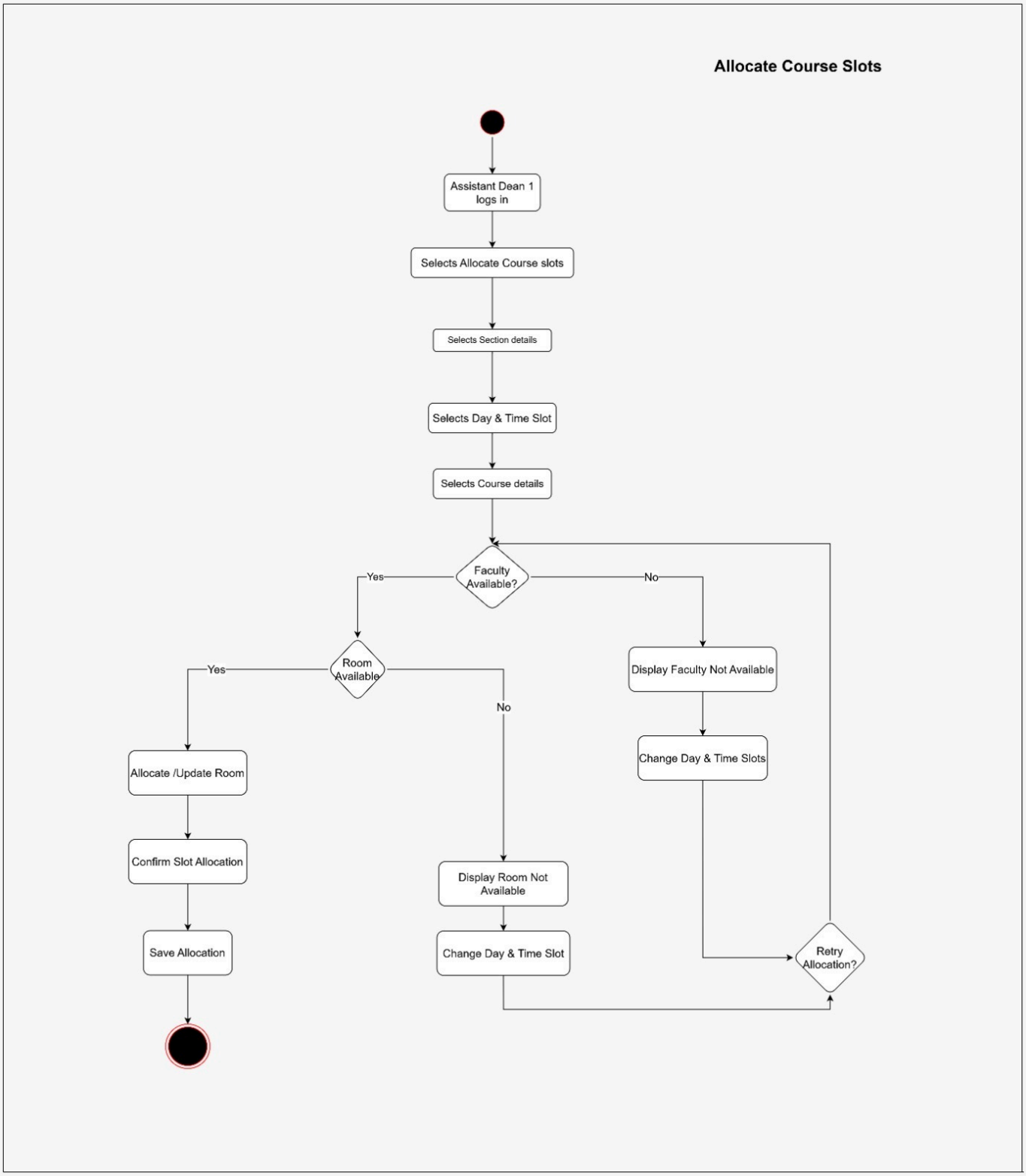
Use Case Diagram

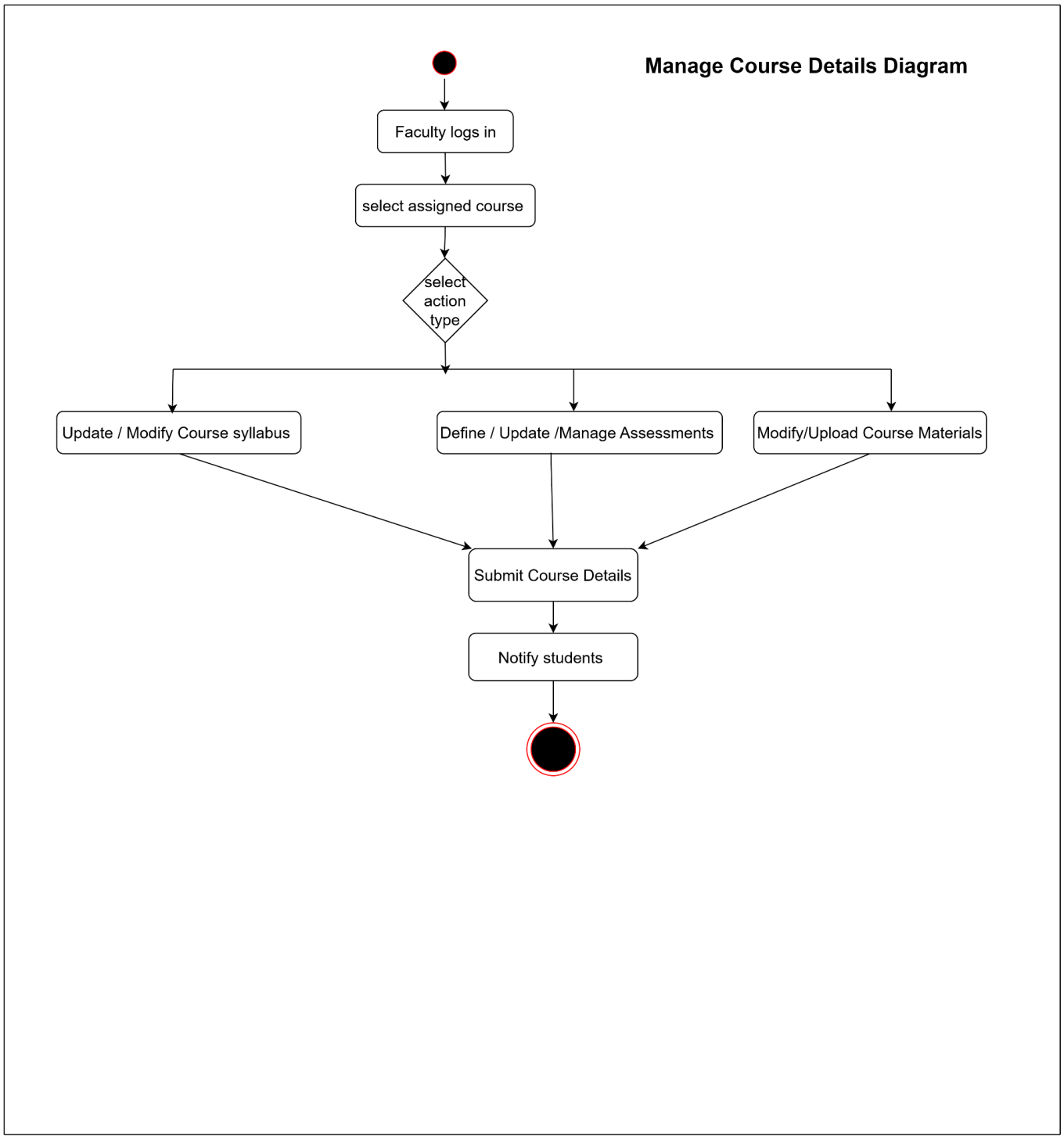



Generate Academic Roadmap Diagram
Select Generate Academic Roadmap
Retrieve Academic History
Execute Degree Audit
Select preferred courses
Pre Requisites Met
Yes
No
Check credit limits
Prerequisite error
Reselect preferred Courses
WithIn The Credit Limits
Yes
No
Build academic roadmap
Suggest alternative courses
Reselect preferred Courses
Save/Export roadmap

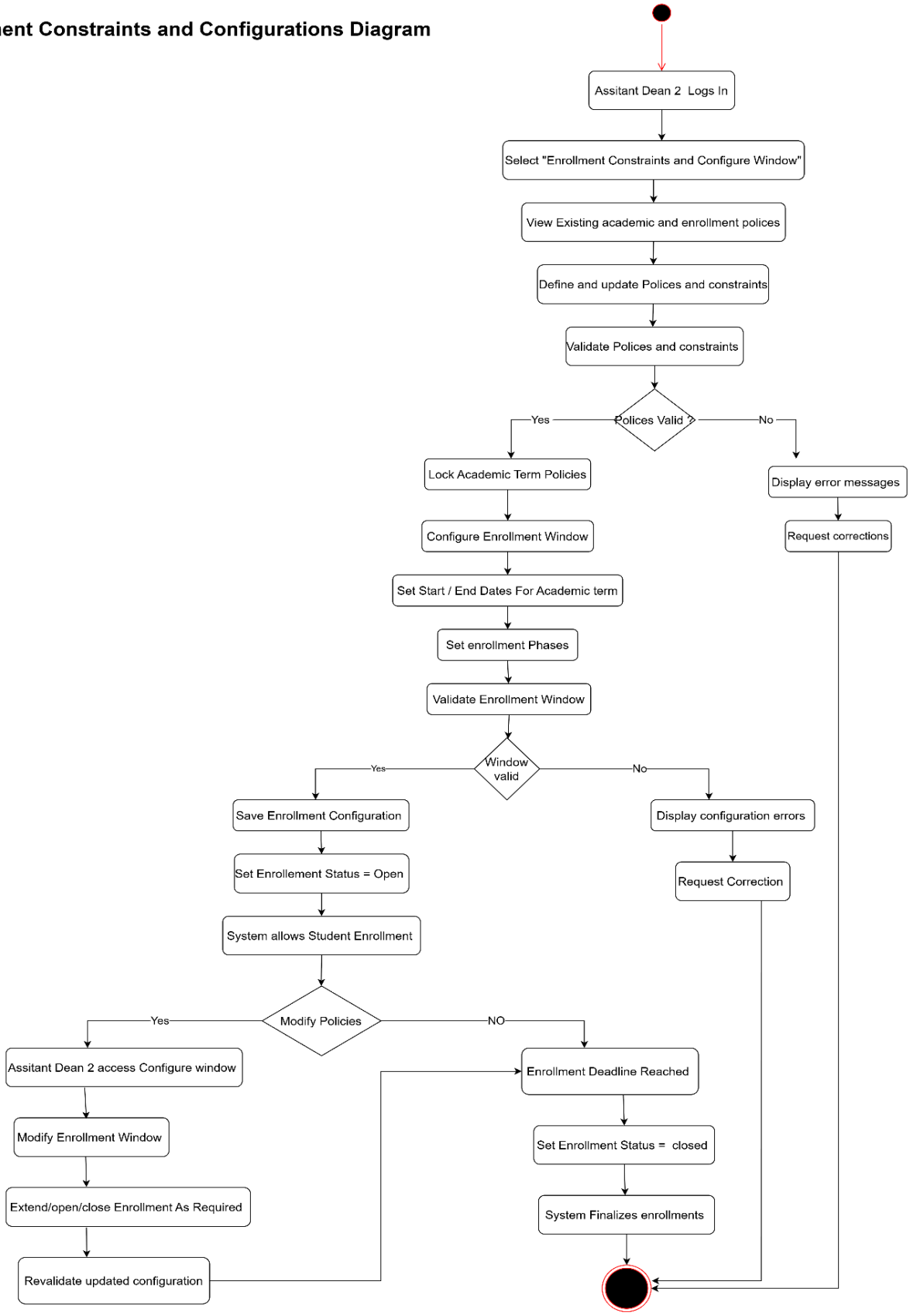

Enroll In Course Diagram
Student starts enroll in Course
Checks enrollment Constraints
Constraints Valid
Yes
No
Enroll Student Succesfully
Display Enrollment Error
Student Requests Override
Yes
No
Student fills Override Request Form
Student Cancels Enrollment Attempt
Student Submits Override Request
Assitant Dean 2 Approved
Yes
No
Enroll Student Successfuly
Notify Student Of Rejection

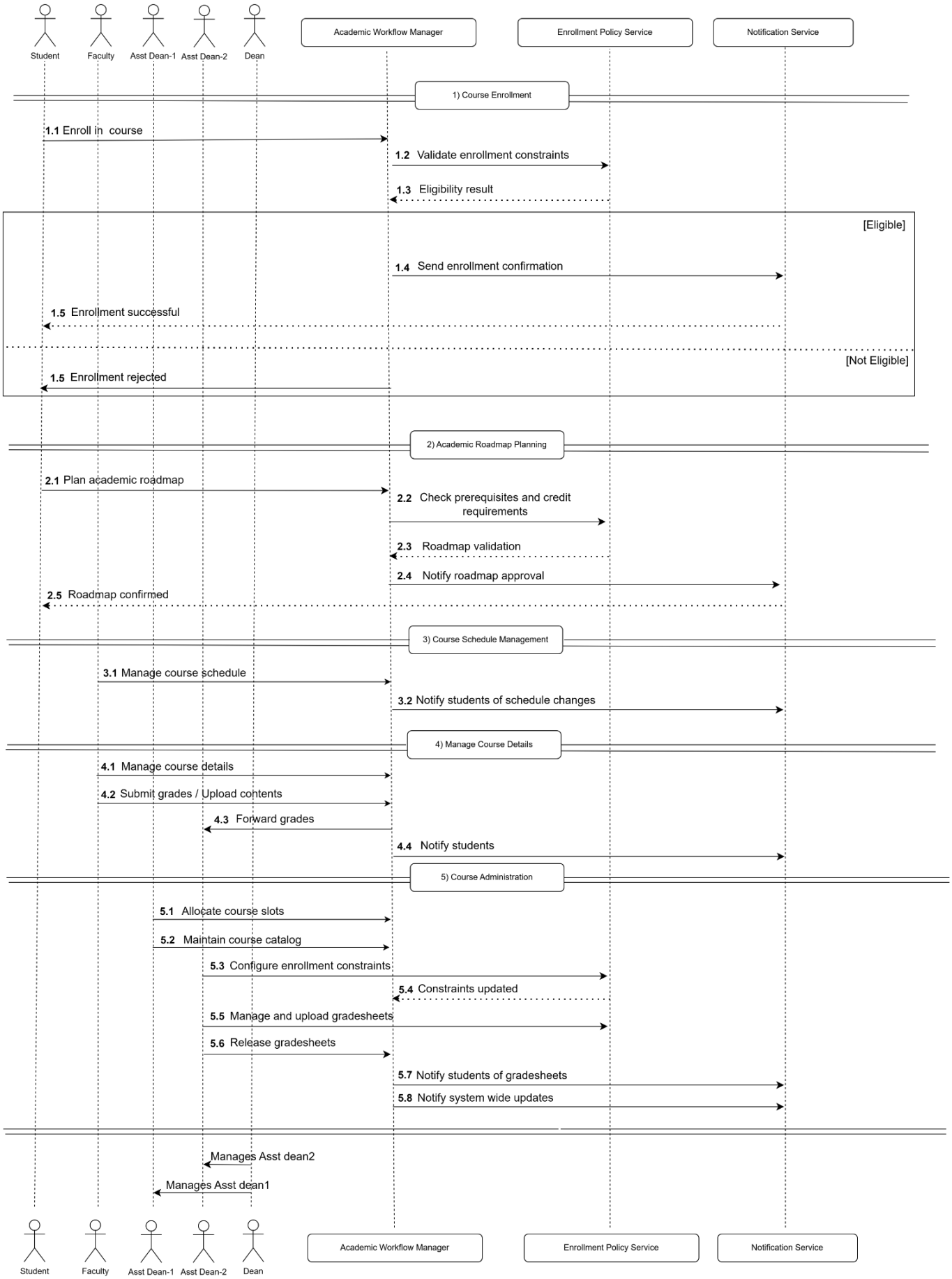

Manage Course Catalog Diagram
Assistant Dean1 logs in
Select mange course catalog
views existing course list
select action
Add course
Update Existing Course
Enter Course detailes
Validate New course Details
New Course Valid Data
Yes
No
Add Course catalog
Display Error Message
Re Enter Course Details
Select existing course
Modify Course Details
Validate Course Details
Updated data Valid
Yes
No
Update Course Catalog
Display error Message
Correct course Details
Deactivate course
Select course To deactivate
Check Course Dependencies
Deactivation Allowed
Yes
No
Mark course Inactive
Display warning Message
Update in Course catalog


Allocate Course Slots
Assistant Dean 1 logs in
Selects Allocate Course slots
Selects Section details
Selects Day & Time Slot
Selects Course details
Faculty Available?
Yes
No
Room Available
Yes
No
Display Faculty Not Available
Allocate /Update Room
Change Day & Time Slots
Confirm Slot Allocation
Display Room Not Available
Save Allocation
Change Day & Time Slot
Retry Allocation?


Manage Course Details Diagram
Faculty logs in
select assigned course
select action type
Update / Modify Course syllabus
Define / Update /Manage Assessments
Modify/Upload Course Materials
Submit Course Details
Notify students

Enrollment Constraints and Configurations Diagram




Student
Faculty
Asst Dean-1
Asst Dean-2
Dean
Academic Workflow Manager
Enrollment Policy Service
Notification Service
1) Course Enrollment
1.1 Enroll in course
1.2 Validate enrollment constraints
1.3 Eligibility result
[Eligible]
1.4 Send enrollment confirmation
1.5 Enrollment successful
[Not Eligible]
1.5 Enrollment rejected
2) Academic Roadmap Planning
2.1 Plan academic roadmap
2.2 Check prerequisites and credit requirements
2.3 Roadmap validation
2.4 Notify roadmap approval
2.5 Roadmap confirmed
3) Course Schedule Management
3.1 Manage course schedule
3.2 Notify students of schedule changes
4) Manage Course Details
4.1 Manage course details
4.2 Submit grades / Upload contents
4.3 Forward grades
4.4 Notify students
5) Course Administration
5.1 Allocate course slots
5.2 Maintain course catalog
5.3 Configure enrollment constraints
5.4 Constraints updated
5.5 Manage and upload gradesheets
5.6 Release gradesheets
5.7 Notify students of gradesheets
5.8 Notify system wide updates
Manages Asst dean2
Manages Asst dean1